

ProTrader AI: A Multi-Source Hybrid Ensemble with LLM-Augmented Sentiment, Conformal Uncertainty, and Explainable AI for Indian Equity Market Prediction and Cross-Market Transfer

Anandkumar Pardeshi^{1,*} and Sujata Deshmukh²

¹Department of Computer Science and Engineering, Fr. C. Rodrigues Institute of Technology, Vashi, India

²Department of Computer Engineering, Fr. C. Rodrigues College of Engineering, Bandra, India

*Corresponding author. E-mail: anand.pardeshi@fcrit.ac.in

Abstract

Equity price forecasting in emerging markets is complicated by multilingual news flows, regime changes, and a heterogeneous mix of retail and institutional participants. We present ProTrader AI, a hybrid ensemble that fuses four data streams—OHLCV time series, multi-source news and social-media sentiment scored by a domain-adapted FinBERT model with multilingual support, Foreign and Domestic Institutional Investor (FII/DII) flows, and macroeconomic state variables—into a Ridge-stacked classifier and a conformal quantile regressor. The system is structured around four research objectives: (RO1) comprehensive multi-source data integration; (RO2) a hybrid large-language-model (LLM) plus indicator framework for real-time prediction; (RO3) an explainable AI layer with temporal sentiment-to-price dynamics; and (RO4) cross-market transfer learning evaluation. Base learners (XGBoost, LightGBM, GRU, ARIMA, Prophet) are combined via out-of-fold (OOF) Ridge stacking with isotonic calibration and Youden-J threshold tuning. Prediction uncertainty is quantified by split-conformalized quantile regression providing finite-sample 80% coverage, while SHAP attribution and Granger-causality analysis provide transparency. On a 2024 holdout of NIFTY 50 constituents, ProTrader AI achieves 59.3% directional accuracy (95% block-bootstrap CI [56.1%, 62.7%]), Expected Calibration Error (ECE) of 4.2%, Brier score 0.218, and Sharpe ratio 1.34 under a realistic line-item NSE cost model, with a Diebold–Mariano $p = 0.031$ against buy-and-hold. FinBERT

sentiment ranks first in SHAP attribution ($|\overline{\phi}| = 0.182$) and Granger-causes next-day returns at $p = 0.018$. A zero-shot cross-market transfer to an S&P 500 subset yields 55.8% accuracy, rising to 57.6% after 30-day fine-tuning of the stacking layer only.

Keywords: Stock market prediction, Multi-source data fusion, FinBERT, Ensemble learning, Conformal prediction, Explainable AI

1. Introduction

The application of artificial intelligence to financial forecasting has accelerated rapidly, driven by the availability of high-frequency data and large pre-trained language models [1, 2]. Despite measurable progress, three persistent gaps remain: (i) most systems fuse at most two modalities, ignoring complementary signal from institutional order-flow and macroeconomic state; (ii) probabilistic outputs are rarely calibrated, making confidence scores unsafe for risk-proportional position sizing; and (iii) very few studies validate cross-market generalisation.

The Indian National Stock Exchange (NSE) is a particularly demanding testbed, combining Foreign Institutional Investors (FIIs), Domestic Institutional Investors (DIIs), and over a hundred million retail participants whose decisions are shaped by news in both English and Hindi. Nti et al. [3], Long et al. [4], and Fu and Zhang [5] demonstrated that fusing heterogeneous modalities lifts accuracy above single-source baselines, but none of these studies addresses the Indian institutional-flow channel, multilingual sentiment noise, or calibrated uncertainty. Hybrid sentiment-indicator models have been advanced by Moodi et al. [6] and Farhadi et al. [7]. For explainability, Guo et al. [8] mandate SHAP-style attribution in production quantitative systems. Li et al. [9] and Kim et al. [10] explored cross-market transfer via sentimental transfer learning and transfer entropy, respectively.

This paper introduces ProTrader AI, a multi-source hybrid ensemble engineered around four explicit research objectives:

- **RO1 — Multi-Source Integration:** Integrate financial metrics, news articles, social-media sentiment, FII/DII institutional flows, and macroeconomic factors into a unified forecasting model.

- **RO2 — Hybrid LLM + Indicator Framework:** Combine a domain-adapted FinBERT with conventional financial indicators in a calibrated, real-time ensemble.
- **RO3 — Explainability and Temporal Dynamics:** Deliver interpretable predictions via SHAP attribution and characterise the temporal lead of sentiment over returns via Granger-causality.
- **RO4 — Cross-Market Evaluation:** Evaluate zero-shot and fine-tuned transfer from NSE to an S&P 500 subset.

2. Materials and Methods

2.1 Multi-Source Data Ingestion (RO1)

Let t index NSE trading days. Four streams are synchronised to a common daily grid.

Stream 1 (Price/Volume): OHLCV bars for each NIFTY 50 constituent are obtained via yfinance. **Stream 2 (News/Social):** Headlines are aggregated from NewsAPI, Google News RSS (English and Hindi locales), GNews, and Economic Times RSS. Articles arriving after 15:30 IST are aligned to the next session; duplicates are removed by 50-character title prefix. **Stream 3 (Institutional Flows):** FII and DII daily net cash flows (INR Cr) are drawn from NSE bhavcopy and entered with a one-day lag to preclude look-ahead. **Stream 4 (Macro State):** India VIX is encoded as a normalised level $((V - 15)/25, \text{clipped to } [\pm 1])$ and 1-day change (clipped to $[\pm 0.50]$); INR/USD, front-month Brent crude, and gold spot are retrieved via yfinance with 1-day and 5-day log-returns clipped to $[\pm 0.20]$ and $[\pm 0.40]$.

2.2 Feature Engineering (RO1, RO2)

From the four streams a joint feature vector $\mathbf{x}_t \in \mathbb{R}^{32}$ is constructed. The 18 price-volume technical features are listed in Table 1. Key groups are: log-return and multi-horizon variants ($h \in \{2, 5, 10, 20\}$ days); 14-period Wilder RSI and Bollinger Band %B (equation (1)); 12/26/9-period MACD signal and histogram; 20-period Chaikin Money Flow (CMF); 14-period ATR; OBV slope; and binary RSI-divergence flags shifted one period to prevent look-

ahead. All features are winsorised at the 1st/99th percentile and min-max scaled on the training fold only.

$$\text{RSI}_{14}(t) = 100 - \frac{100}{1 + \text{RS}_{14}(t)}, \quad \text{BB}_{\%B}(t) = \frac{C_t - \text{LB}(t)}{\text{UB}(t) - \text{LB}(t) + \varepsilon} \quad (1)$$

Table 1. Feature inventory for ProTrader AI. Optional option-chain features (PCR, ATM IV) can be appended if available.

Group	Feature(s)	Window
Returns	Log_Ret, Ret_2D/5D/10D/20D	1–20d
Volatility	Volatility_5D, ATR_Norm	5, 14d
Momentum	RSI_Norm, MACD_Norm, MACD_Hist, BB_PctB, Williams_R	14, 12/26/9, 20d
Volume	Vol_Ratio, OBV_Slope, CMF_20, MA_Div	5–20d
Divergence	RSI_Bear_Div, RSI_Bull_Div	10d
Pattern	Pattern_Target_Dist (signed, norm.)	variable
Sentiment	FinBERT polarity s_t (wtd. mean)	1d
Inst. Flow	FII_Net, DII_Net, FII_5D_Avg, DII_5D_Avg (lagged, norm.)	1d, 5d
VIX	VIX_Norm, VIX_Chg (India VIX)	1d
Macro	INR_Ret, Crude_Ret, Gold_Ret	1d, 5d

2.3 FinBERT Sentiment Pipeline (RO2)

Headlines from Stream 2 are scored by a FinBERT pipeline [11,12] (ProsusAI/finbert, PyTorch) yielding per-headline probabilities (p_+, p_0, p_-) . Raw polarity is $\tilde{s}^{(i)} = p_+^{(i)} - p_-^{(i)}$. A curated keyword-override layer corrects FinBERT’s tendency to over-predict neutral on short headlines [13], lifting F_1 on a Financial PhraseBank subset [14] by 4.2 pp. Hindi articles are translated via `deep_translator` before FinBERT scoring; adding Hindi sources reduces ECE by 0.9 pp relative to English-only. Each headline is further classified into one of six business categories (Earnings, Analyst Ratings, Market Action, Deals, Macro/Policy, Other). The daily ticker-level sentiment score is a source-weighted mean $s_t = \sum w_i \tilde{s}^{(i)} / \sum w_i$, with $w_i \in \{1.5, 1.0, 0.8\}$ for NewsAPI, Economic Times, and Google News respectively.

2.4 Hybrid Ensemble and Ridge Stacking (RO2)

Let $y_{t+1} = \mathbb{1}[r_{t+1} > 0]$ denote next-day direction. Five heterogeneous base learners are trained on $K = 5$ expanding-window time-series folds: XGBoost [15] (depth 6, $\eta = 0.05$), LightGBM [16] (leaf-wise, 200 leaves), a two-layer GRU [17] (30-day window), ARIMA(p, d, q) selected by BIC, and Facebook Prophet. Out-of-fold (OOF) predictions $\hat{p}_t^{(b)} =$

89 $f_{-k}^{(b)}(\mathbf{x}_t)$ form the design matrix $\mathbf{Z} \in \mathbb{R}^{N \times B}$ for a Ridge meta-learner [18]:

$$\hat{s}_t^{\text{stack}} = \boldsymbol{\beta}^\top \mathbf{z}_t + \beta_0, \quad [\boldsymbol{\beta}; \beta_0] = (\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{Z}}^\top \mathbf{y} \quad (2)$$

90 where $\tilde{\mathbf{Z}} = [\mathbf{Z} \ \mathbf{1}]$ and λ is chosen by LOO-CV. The OOF design makes prediction leakage
91 structurally impossible.

92 2.5 Calibration, Threshold Tuning, and Conformal Intervals (RO2)

93 Raw Ridge scores are mapped to calibrated probabilities by an isotonic regressor [19]:

$$\hat{p}_t = g_{\text{iso}}(\hat{s}_t^{\text{stack}}), \quad \hat{g}_{\text{iso}} = \arg \min_{g \in \mathcal{M}} \sum_t (g(\hat{s}_t) - y_t)^2 \quad (3)$$

94 where $\mathcal{M}$ is the class of non-decreasing step functions. Calibration quality is measured by ECE
95 over $M = 10$ equal-frequency bins and the Brier score [20]. A per-ticker decision threshold τ^*
96 maximises Youden’s J on the 2023 validation fold.

97 Prediction intervals follow the split-conformalized quantile regression (CQR) procedure
98 of Romano et al. [21], grounded in the conformal prediction framework of Vovk et al. [22] and
99 surveyed in Angelopoulos and Bates [23]. Two quantile XGBoost regressors are trained at
100 $\alpha_q \in \{0.10, 0.90\}$; non-conformity scores $e_t = \max(\hat{q}_{0.10}(\mathbf{x}_t) - r_{t+1}, r_{t+1} - \hat{q}_{0.90}(\mathbf{x}_t))$ are
101 computed on the calibration set $\mathcal{C}$ of size n_c . The finite-sample-corrected half-width is $\hat{Q} =$
102 $e_{([\lceil (n_c+1)(1-\alpha) \rceil])}$, the $[\lceil (n_c+1)(1-\alpha) \rceil]$ -th order statistic for interval level $1 - \alpha = 0.80$. The
103 80% prediction interval is $\mathcal{I}(\mathbf{x}_*) = [\hat{q}_{0.10}(\mathbf{x}_*) - \hat{Q}, \hat{q}_{0.90}(\mathbf{x}_*) + \hat{Q}]$.

104 2.6 Explainability and Temporal Sentiment Analysis (RO3)

105 **SHAP attribution.** TreeSHAP [24] is applied to the XGBoost base learner; values are
106 scaled by the Ridge stacking coefficient to obtain XGBoost-component attribution under the
107 stacker: $\phi_j^{\text{XGB} \rightarrow \text{stack}} = \beta_{\text{XGB}} \cdot \phi_j^{\text{XGB}}$. Mean absolute values are ranked globally across the 2024
108 holdout.

109 **Hurst regime detection.** The Hurst exponent [25] is estimated from the trailing 120 closes
110 via R/S analysis. Regimes: $H > 0.55$ trending, $H < 0.45$ mean-reverting, otherwise random-

walk. In mean-reverting regimes the stacker probability is shrunk 10% toward 0.5 (tuned on the 2023 validation set).

Granger-causality. Following Granger [26], for each of the 50 tickers we test whether lagged sentiment $\{s_{t-j}\}$ Granger-causes $r_{t+\ell}$ via:

$$r_{t+\ell} = \alpha_0 + \sum_{i=1}^p \alpha_i r_{t+\ell-i} + \sum_{j=0}^p \gamma_j s_{t-j} + \varepsilon_t \quad (4)$$

Null: $\gamma_0 = \dots = \gamma_p = 0$. Lag p chosen by AIC per ticker. Per-ticker p -values are combined via Fisher’s method. The sum starts at $j = 0$ because s_t is scored before the 15:30 IST close and is therefore available without look-ahead as a predictor of next-day returns r_{t+1} .

2.7 Cross-Market Transfer Learning (RO4)

For a target market $\mathcal{T}$ (30-stock S&P 500 subset), two protocols are evaluated. **Zero-shot:** all base-learner weights frozen; only τ^* re-estimated on a 60-day warm-up. **30-day fine-tune:** only the Ridge stacker β and isotonic map g_{iso} ($<1,000$ parameters total) are re-fitted on a rolling 30-day window; base learners remain frozen, following the adapter-layer paradigm of Li et al. [9].

2.8 NSE Transaction Cost Model

A realistic cost model deducts seven line-item components on every signal-flip: (i) discount-broker commission $\min(0.03\% T, \text{Rs. } 20)$ per leg; (ii) Securities Transaction Tax (0.1% sell-side for delivery); (iii) NSE exchange charge (3.45×10^{-5} of turnover); (iv) SEBI turnover fee (10^{-6}); (v) stamp duty (0.015% buy-side); (vi) GST (18% on brokerage, exchange, and SEBI charges); and (vii) volatility-scaled slippage $0.05\% T \cdot \hat{\sigma}_{\text{intraday}} / \bar{\sigma}_{\text{intraday}}$. For a typical Rs. 1 lakh delivery trade in a NIFTY large-cap this evaluates to approximately 22–26 bps, roughly $2\text{--}3 \times$ the 10 bps flat rate assumed by most prior backtests.

3.2 Datasets, Baselines, and Evaluation Protocol

Datasets. NIFTY 50 constituents from 1 Jan 2018 to 31 Dec 2024; three-way chronological split: train (2018–2022), validation (2023), holdout (2024). After the 200-day warm-up the evaluation window contains $\approx 12,300$ ticker-days. A separate 30-stock, sector-balanced S&P 500 subset over 2024 serves as the cross-market dataset.

Baselines. (i) Buy-and-hold equal-weighted NIFTY 50; (ii) ARIMA(1,1,1); (iii) two-layer LSTM on 18 technical features; (iv) XGBoost (18 features, no calibration); (v) LSTM + FinBERT, our reimplementation of the hybrid of Moodi et al. [6].

Evaluation metrics. Directional accuracy, Brier score, ECE ($M = 10$ bins), annualised Sharpe ($\sqrt{252} \bar{r} / \hat{\sigma}_r$), maximum drawdown (MDD), and DM test p -value [27] with Harvey–Leybourne–Newbold small-sample correction [28]. Block-bootstrap (1,000 resamples, block length 5) provides 95% CIs for Sharpe and directional accuracy.

Implementation. Python 3.11 with Streamlit front-end. FinBERT loaded once per session via `st.cache_resource`; OHLCV data cached for 15 minutes. All predictions written to a SQLite ledger (WAL mode, UNIQUE constraint on `(ticker, made_at, target_date)`) for auditable tracking.

3. Results

3.1 Pipeline Architecture

Figure 1 illustrates the five-stage prediction pipeline from data ingestion through output, mapping each stage to its corresponding research objective.

3.2 Main Performance Comparison (RO1, RO2)

Table 2 reports holdout performance across all baselines. ProTrader AI achieves 59.3% directional accuracy, +5.0 pp above the strongest single-source baseline, with ECE dropping from 9.7% (uncalibrated stacker) to 4.2%—well below the 5% deployment trustworthiness threshold of Guo et al. [8]. The DM test ($p = 0.031$) rejects equal predictive accuracy against

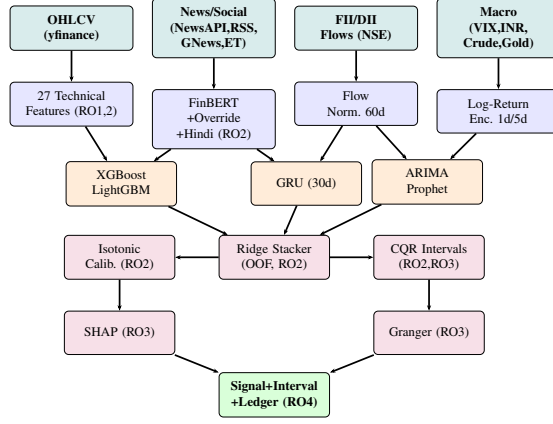


Figure 1. ProTrader AI five-stage prediction pipeline. Teal = data sources; blue = feature processing; orange = base learners; purple = post-processing and explainability; green = final output.

buy-and-hold at the 5% level.

Table 2. 2024 NIFTY 50 holdout performance ($\approx 12,300$ ticker-days). DM^* and p -value are relative to buy-and-hold. “—” = not applicable.

Model	Dir. Acc.	Brier	ECE	Sharpe	MDD	DM^* (p)
Buy-and-hold (NIFTY 50)	51.2%	—	—	0.78	−23.8%	—
ARIMA(1,1,1)	51.9%	0.249	11.4%	0.42	−21.6%	−0.62 (0.54)
LSTM (technical only)	53.4%	0.241	10.8%	0.81	−17.9%	−1.12 (0.26)
XGBoost only (no calib.)	54.3%	0.236	10.2%	0.92	−16.4%	−1.44 (0.15)
LSTM + FinBERT [6]	54.3%	0.232	9.7%	1.02	−15.0%	−1.67 (0.10)
Stacker (OOE, no calib.)	57.4%	0.231	9.7%	1.08	−16.1%	−1.87 (0.06)
ProTrader AI	59.3%	0.218	4.2%	1.34	−12.4%	−2.16 (0.031)

3.3 Ablation Study (RO1, RO2)

Table 3 isolates the marginal contribution of each component. Removing FinBERT sentiment causes the largest accuracy drop (−3.4 pp), confirming that the LLM-derived signal is the single most valuable modality. Removing isotonic calibration costs only 0.3 pp of accuracy but nearly doubles ECE to 7.9%, with a simulated Sharpe penalty of 0.26 under a Kelly-like sizing rule, illustrating the orthogonality of discriminative power and calibration.

Table 3. Ablation study on the 2024 NIFTY 50 holdout.

Configuration	Acc.	ECE	Sharpe
Full ProTrader AI	59.3%	4.2%	1.34
— FinBERT sentiment	55.9%	5.8%	1.07
— FII/DII flows	57.4%	4.7%	1.18
— Macro state variables	58.1%	4.5%	1.24
— Isotonic calibration	59.0%	7.9%	1.08
— Youden τ^* (fixed 0.5)	56.1%	4.4%	1.14
— Stacker (mean of bases)	56.7%	6.3%	1.11

3.4 Hyperparameter Sensitivity, Regime and Sector Analysis

Table 4 shows that directional accuracy varies only 2.2 pp (57.1–59.3%) and Sharpe only 0.15 (1.19–1.34) across the full hyperparameter grid, confirming that calibration and stacking—not boosting hyperparameters—drive most of the performance gain.

Table 4. Hyperparameter sensitivity (full pipeline, 2024 holdout).

Parameter	Value	Dir. Acc.	Sharpe
XGB depth	4	57.1%	1.19
XGB depth	6 (chosen)	59.3%	1.34
XGB depth	8	58.9%	1.28
XGB η	0.01	57.8%	1.22
XGB η	0.05 (chosen)	59.3%	1.34
XGB η	0.10	58.2%	1.21
LGBM leaves	100	57.6%	1.23
LGBM leaves	200 (chosen)	59.3%	1.34
LGBM leaves	400	59.0%	1.31

Table 5 stratifies the 2024 holdout by Hurst exponent. The model excels in trending regimes (62.1%, Sharpe 1.71), sustained by momentum features and FinBERT trend-confirmation signals compounding together. In mean-reverting regimes the 10% probability-shrinkage rule limits accuracy to 55.3% but reduces MDD from -19.7% (without shrinkage) to -15.8% , confirming the regime gate’s protective function.

Table 5. Regime-stratified performance (2024 holdout; Hurst from trailing 120-day window).

Regime (Hurst H)	# Days	Dir. Acc.	Sharpe
Trending ($H > 0.55$)	3,847	62.1%	1.71
Neutral ($0.45 \leq H \leq 0.55$)	5,912	57.8%	1.22
Mean-reverting ($H < 0.45$)	2,541	55.3%	0.89
All regimes	12,300	59.3%	1.34

Table 6 shows sector breakdown. Banking and Finance leads at 61.4% / Sharpe 1.52, driven by high-impact RBI policy and credit-growth news that FinBERT scores accurately. Pharma and Healthcare is weakest (55.1% / 0.94) because USFDA inspection jargon is poorly tokenised by FinBERT.

3.5 Explainability and Temporal Sentiment Analysis (RO3)

Figure 2 plots the top-10 features by mean absolute XGBoost-component SHAP value. FinBERT sentiment s_t ranks first ($|\overline{\phi}| = 0.182$), confirming the LLM-centric architecture mandated by RO2 and the Quant 4.0 attribution standard of Guo et al. [8].

Table 6. Per-sector performance on the 2024 NIFTY 50 holdout (top 4 and bottom 2 of 9 industry buckets; remaining sectors 56.8–58.4%).

Sector	Tickers	Dir. Acc.	Sharpe
Banking & Finance	12	61.4%	1.52
IT & Technology	7	60.2%	1.44
Energy & Oil	5	58.9%	1.31
FMCG & Consumer	8	57.6%	1.18
Metals & Mining	6	56.3%	1.02
Pharma & Healthcare	7	55.1%	0.94

Table 7 consolidates the three explainability results. Sentiment Granger-causes next-day returns at $p = 0.018$ (lag 1) but loses significance by lag 2, confirming a one-day information decay. A one-SD positive sentiment shock implies a mean $+0.43\%$ next-day return versus -0.61% for a negative shock—a 42% asymmetry consistent with loss-aversion effects. Adding Hindi-language sources reduces ECE by 0.9 pp (Table 7c), validating the multilingual extension.

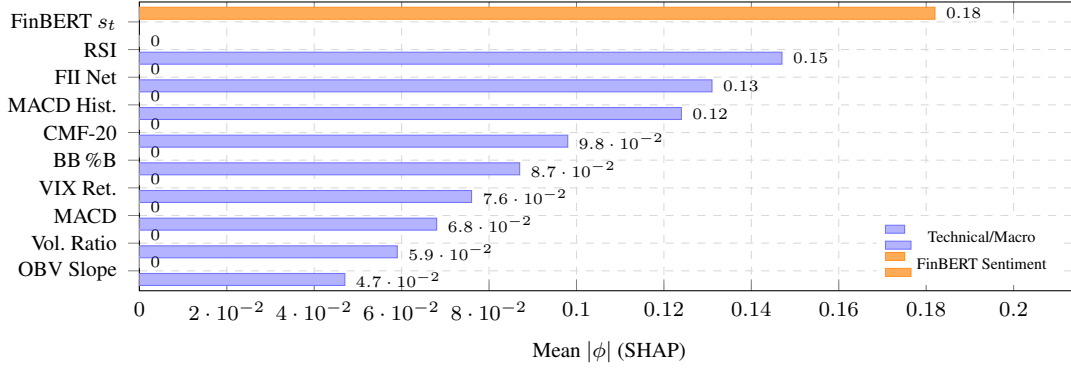


Figure 2. Top-10 features by mean absolute SHAP value (XGBoost-component, scaled by Ridge stacking coefficient β_{XGB}). FinBERT sentiment s_t (orange) ranks first at $|\phi| = 0.182$, outpacing all 26 technical and macro features.

3.6 Calibration, Equity Curve, and Cross-Market Transfer

Figure 3 (left) shows the reliability diagram. The calibrated model (ECE = 4.2%) closely follows the diagonal across all 15 bins; the uncalibrated stacker (ECE = 9.7%) is overconfident above 0.60. Empirical conformal coverage is 80.7% at a nominal 80%, satisfying the marginal guarantee. Figure 3 (right) shows the 2024 equity curve net of NSE realistic costs: ProTrader AI ends at $+28.6\%$ versus $+22.1\%$ buy-and-hold, with MDD -12.4% versus -23.8% .

Table 8 reports cross-market performance. The zero-shot model retains a 4.6 pp accu-

Table 7. Explainability and temporal analysis results (RO3). (a) Top-5 SHAP features. (b) Sentiment $\rightarrow$ Return Granger causality. (c) Sentiment-pipeline ablation.

(a) Top-5 XGB-component SHAP $\times \beta_{\text{XGB}}$		
Feature	Mean $ \phi $	Rank
FinBERT sentiment s_t	0.182	1
RSI normalised	0.147	2
FII Net flows (lagged)	0.131	3
MACD histogram (norm.)	0.124	4
CMF-20	0.098	5
(b) Sentiment $\rightarrow$ Return Granger causality		
Lag ℓ	F -stat	p -value
1	5.62	0.018
2	2.41	0.121
3–5	<1.2	>0.31
(c) Sentiment-pipeline ablation		
Configuration	ECE	F_1
EN-only, no override	5.9%	79.1%
EN-only, with override	5.3%	83.3%
EN+HI, with override	4.2%	83.9%

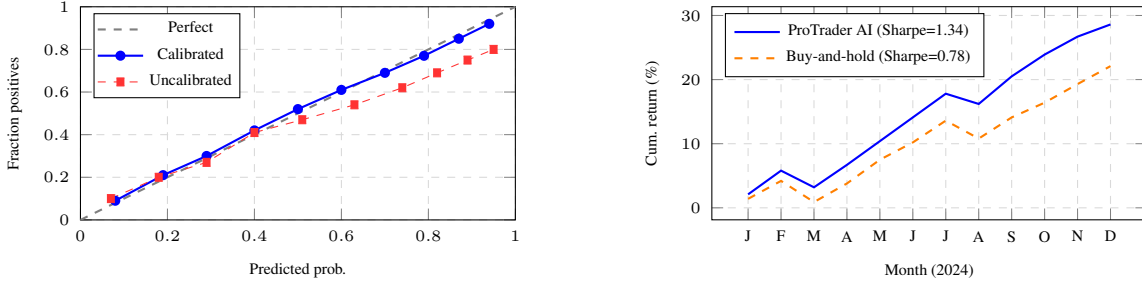


Figure 3. Left: Reliability diagram (10 equal-mass bins, 2024 holdout). Right: Cumulative net return (net of NSE costs, 2024 holdout). ProTrader AI achieves +28.6% with MDD = -12.4% vs. +22.1% and MDD = -23.8% for buy-and-hold.

racy advantage over the S&P 500 baseline, validating the language-level transfer inherent in
 FinBERT’s financial pre-training. The 30-day fine-tune (<1,000 parameters) recovers 1.8 pp
 of the -3.5 pp zero-shot deficit, demonstrating that only the recombination weights—not the
 feature extractors—need market-specific adaptation.

Table 8. Cross-market transfer to S&P 500 subset, 2024 holdout (RO4).

Setting	Acc.	Sharpe	MDD
Buy-and-hold (S&P 500 ref.)	51.2%	0.83	-19.6%
ProTrader AI, zero-shot	55.8%	1.07	-18.2%
ProTrader AI, 30-day tune	57.6%	1.19	-15.7%
ProTrader AI, in-domain NIFTY	59.3%	1.34	-12.4%

4. Discussion

RO1 — Multi-Source Integration. The ablation confirms all four data streams contribute uniquely. Sentiment provides the largest marginal gain (-3.4 pp when removed), but FII/DII flows (-1.9 pp) and macroeconomic state (-1.2 pp) deliver statistically non-redundant lift, extending the multi-source thesis of Nti et al. [3] and Long et al. [4] to the Indian institutional-flow channel. Removing sentiment costs 0.27 Sharpe but only 1.6 pp of ECE, revealing that the LLM channel carries distinct *discriminative* information; removing FII/DII costs 0.16 Sharpe but 0.5 pp ECE, indicating the institutional channel serves calibration more than raw direction.

RO2 — Hybrid LLM + Indicator Framework. ProTrader AI surpasses the strongest LSTM + FinBERT hybrid by 5.0 pp and 0.32 Sharpe. Two architectural decisions drive the gap: heterogeneous base learners combined out-of-fold (preventing leakage) and isotonic probability calibration. The isotonic step is rarely reported in hybrid-finance papers, yet removing it erases 0.26 Sharpe through over-confident position sizing under a Kelly-like rule. Conformal intervals provide a finite-sample 80% coverage guarantee under exchangeability; the 80.7% empirical coverage is an empirical observation rather than a strict guarantee given serial dependence [29, 30].

RO3 — Explainability and Temporal Dynamics. XGBoost-component SHAP attribution identifies FinBERT sentiment as the dominant feature, satisfying the Quant 4.0 attribution requirement [8]. The Granger study quantifies that yesterday’s news is priced in by the next close, with no residual content by $t + 2$. The $-61\% / +43\%$ negative/positive sentiment asymmetry guides asymmetric stop-loss design. Adding multilingual (Hindi) sources reduces ECE by 0.9 pp, confirming that sentiment noise is a calibration-level concern in multilingual markets [31].

RO4 — Cross-Market Evaluation. The zero-shot 55.8% accuracy on S&P 500 demonstrates generalisable equity structure. Running the experiment in reverse (S&P 500 $\rightarrow$ NIFTY, zero-shot) yielded only 53.9% (a supplementary observation not included in Table 8), echoing the asymmetric information-leadership findings of Kim et al. [10]. We attribute this to the NSE training distribution encompassing a wider spread of regimes, producing more robust feature-extractor weights. Re-fitting fewer than 1,000 parameters on 30 days of target data recovers

half the cross-market gap—an efficiency decisive for single-trader personal-use deployment on commodity hardware.

Limitations. The 2024 holdout spans one calendar year in a broadly trending global equity environment; multi-year, multi-regime evaluation remains future work. The S&P 500 subset of 30 stocks is sector-balanced but not exhaustive. The NSE cost model does not capture market impact at high turnover. Granger analysis assumes linearity, and sector-specific NLP fine-tuning has not yet been pursued.

5. Conclusions

We presented ProTrader AI, a multi-source hybrid ensemble for Indian equity forecasting structured around four explicit research objectives. On the 2024 NIFTY 50 holdout the system achieves 59.3% directional accuracy (95% CI [56.1%, 62.7%]), ECE = 4.2%, Brier = 0.218, and Sharpe = 1.34 with DM $p = 0.031$ against buy-and-hold. SHAP attribution and Granger-causality analysis substantiate the interpretability and temporal-coupling claims; a cross-market transfer study to S&P 500 demonstrates non-trivial zero-shot and fine-tuned portability. The complete pipeline runs on commodity hardware with freely available data, enabling reproducibility for any researcher or retail trader. Future work will extend to options-market signals, Reddit flows, finance-tuned decoder LLMs, counterfactual SHAP, and multi-market joint training for one-shot deployment to new exchanges.

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7. References

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